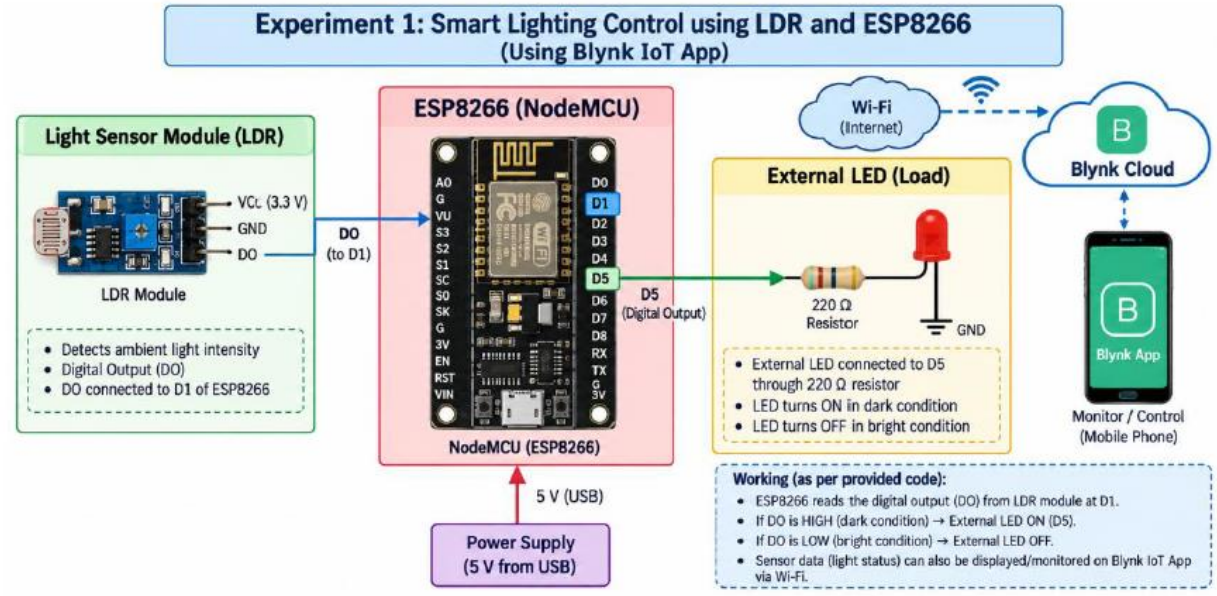
ASSIGNMENT – 5

BATCH 8

Aim: Read **Light Intensity** using **LDR and ESP8266** and display the sensor data on the **Blynk IoT App**.

Circuit Diagram:



1. Simple LDR Experiment

Code:

```
// Smart Lighting Control using ESP8266
// LDR Module: VCC, GND, DO
// LDR Module DO -> D1
// External LED -> D5 through 220 ohm resistor
#define LDR_PIN D1
#define LED_PIN D5

void setup() {

  Serial.begin(115200);
  pinMode(LDR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);
  Serial.println("Smart Lighting System Started");

 }
```

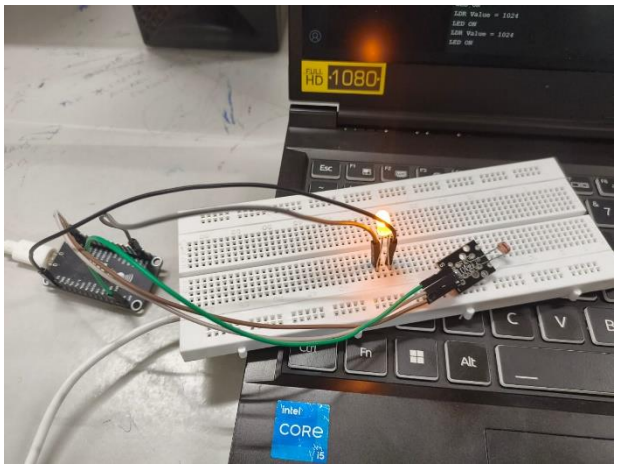
```
void loop() {
        int ldrState = digitalRead(LDR_PIN);
        Serial.print("LDR DO = ");
        Serial.print(ldrState);
        // DARK condition
        // Change HIGH/LOW here if your module gives opposite DO
        if (ldrState == HIGH) {
                digitalWrite(LED_PIN, HIGH);
                Serial.println(" -> DARK: External LED ON");
        }
        else {
                // BRIGHT condition
                digitalWrite(LED_PIN, LOW);
                Serial.println(" -> DARK: External LED ON");
        }
        else {
                // BRIGHT condition
                digitalWrite(LED_PIN, LOW);
                Serial.println(" -> BRIGHT: External LED OFF");

        }

        delay(500);

        }
```

Output:



2. LDR Experiment using Blynk

Code:

```
#define BLYNK_TEMPLATE_ID "TMPL3qD-QJFa9"
#define BLYNK_TEMPLATE_NAME "Street Light"
#define BLYNK_AUTH_TOKEN "r8YSrNFcNq2KNLxHfQdlydnUXa5KnboP"
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
// Wi-Fi credentials
char ssid[] = "GITAM-5GHz";
char pass[] = "Gitam$$123";
// Pin connections
#define LDR_PIN D1
#define LED_PIN D5
BlynkTimer timer;

void checkLight() {
        int ldrState = digitalRead(LDR_PIN);
        Serial.print("LDR DO = ");
        Serial.print(ldrState);
        // YOUR MODULE:
        // HIGH = DARK
         // LOW  = BRIGHT
         if (ldrState == HIGH) {
                  // DARK
                  digitalWrite(LED_PIN, HIGH);
                  Blynk.virtualWrite(V0, 1);
                  Serial.println(" -> DARK -> STREET LIGHT ON");
        }
         else {
                // BRIGHT
                  digitalWrite(LED_PIN, LOW);
                  Blynk.virtualWrite(V0, 0);
                  Serial.println(" -> BRIGHT -> STREET LIGHT OFF");
        }
}
```

```
void setup() {
        Serial.begin(115200);
        pinMode(LDR_PIN, INPUT);
        pinMode(LED_PIN, OUTPUT);
        digitalWrite(LED_PIN, LOW);
        Serial.println();
        Serial.println("================================");
        Serial.println("SMART STREET LIGHTING SYSTEM");
        Serial.println("================================");
      Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
      // Check LDR every 1 second
      timer.setInterval(1000L, checkLight);
}

void loop() {
      Blynk.run();
      timer.run();
}
```

Output:

